A2 · Unit 4 · WPH14 | Further Mechanics, Fields & Particles

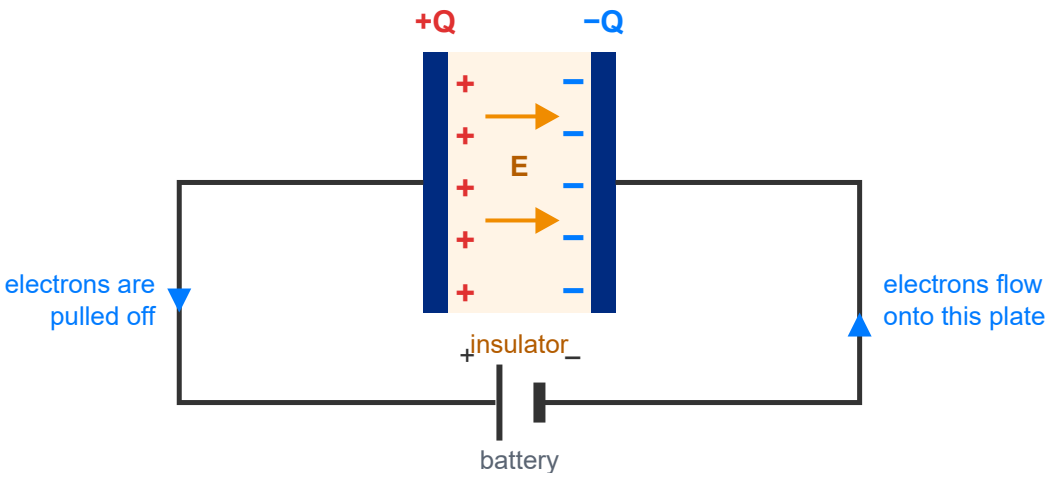
Capacitors

A capacitor stores energy by holding positive and negative charge apart. It stores far less energy than a battery, but it can hand all of it back in a fraction of a second, which is why camera flashes, defibrillators and the smoothing circuit in every phone charger use one. On this page you will see where the ½ in $W = \frac{1}{2}QV$ comes from, why capacitors charge and discharge exponentially, and how a log graph turns that curve into a straight line.

SPEC 100 Capacitance: C = Q/V

WHAT THIS SPEC POINT ASKS

Capacitance is defined as C = Q/V; use this equation.



Charging: the battery moves electrons round the circuit from the left plate to the right plate. The plates end up with equal and opposite charges, and an electric field across the gap.

What a capacitor does

- A capacitor is two conducting plates separated by an insulator (the **dielectric**).
- When connected to a supply, electrons are pushed onto one plate and pulled off the other. One plate gets charge **+Q**, the other **−Q**. The **total** charge on the capacitor is zero; it **separates** charge.
- "Charge stored Q" means the charge on **one** plate.
- The p.d. across the plates rises as charge builds up. Q is **proportional** to V.

Capacitance

FORMULA SHEET

$$C = \frac{Q}{V}$$

C: capacitance / F Q: charge on one plate / C V: p.d. across the capacitor / V

- C = capacitance (farad, F), Q = charge on one plate (C), V = p.d. across the capacitor (V). 1 F = 1 C V^{-1}.
- A farad is huge. Typical values: μF (10^{-6} F), nF (10^{-9} F), pF (10^{-12} F).
- On a graph of **Q against V**, the gradient is C. On a graph of **V against Q**, the gradient is **1/C**.

WORKED EXAMPLE

charge stored

A 470 μF capacitor is charged to 9.0 V.

Q = CV = 470 × 10^{-6} × 9.0 = **4.2 × 10^{-3} C**

MARK-SCHEME WORDING

- Capacitors "separate charge"; "the total charge on a capacitor is zero".
- Sharing: when a charged capacitor is connected to an uncharged one, "the charge stored in each is **less** than the original, because the **p.d. is shared**".

Often asked as: Q = CV after finding C from a time constant; initial current V/R; MCQ on the gradient of a V–Q or Q–V graph.

COMMON MISTAKE

Mixing up C (capacitance, in F) with C (the coulomb, unit of charge). Write units carefully.

Answering a sharing question in terms of "capacitance changes" when it is the p.d. that is shared.

SPEC 101 Energy stored by a capacitor

WHAT THIS SPEC POINT ASKS

Use W = ½QV; derive it from the area under a p.d.–charge graph; derive and use W = ½CV^2 and W = ½Q^2/C.

Derivation from the V–Q graph

- While charging, the p.d. across the capacitor rises in proportion to the charge: the graph of V against Q is a **straight line through the origin** (gradient 1/C).
- To add a small charge ΔQ at p.d. V, work done = VΔQ: a thin strip under the graph.
- Adding all the strips: **energy stored = area under the V–Q graph** = area of a triangle = ½ × Q × V.

Energy stored in a capacitor

FORMULA SHEET

$$W = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2 = \frac{1}{2}\frac{Q^2}{C}$$

W: energy stored / J Q: charge / C V: p.d. / V C: capacitance / F

Deriving W = ½QV from the area under the V–Q graph

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. To move a small extra charge ΔQ onto the plates when the p.d. is V, you do work (p.d. is work done per unit charge):

$$\Delta W = V\,\Delta Q$$

If ΔQ is small, V hardly changes while you add it.

2. On a graph of V (vertical) against Q (horizontal), $V\,\Delta Q$ is the area of a thin strip under the line. The total work done charging from 0 to Q is the sum of all the strips, which is the area under the graph:

$$W = \sum V\,\Delta Q = \text{area under the } V\text{–}Q \text{ graph}$$

3. Because $V \propto Q$, the graph is a straight line through the origin, so the area is a triangle with base Q and height V:

$$W = \tfrac{1}{2} \times \text{base} \times \text{height} \quad \Rightarrow \quad \boxed{W = \tfrac{1}{2}QV}$$

The ½ is there because, on average, each coulomb was pushed on against only half of the final p.d.

Deriving W = ½CV² and W = ½Q²/C

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Start from the result above and the definition of capacitance:

$$W = \tfrac{1}{2}QV, \qquad C = \frac{Q}{V}$$

2. To get rid of Q, write $Q = CV$ and substitute:

$$W = \tfrac{1}{2}(CV)V \quad \Rightarrow \quad \boxed{W = \tfrac{1}{2}CV^2}$$

3. To get rid of V instead, write $V = Q/C$ and substitute:

$$W = \tfrac{1}{2}Q\left(\frac{Q}{C}\right) \quad \Rightarrow \quad \boxed{W = \frac{1}{2}\frac{Q^2}{C}}$$

All three are the same energy. Pick the one that uses the two quantities you already know.

- W = energy stored (J), Q = charge (C), V = p.d. (V), C = capacitance (F).
- The other forms come from substituting Q = CV (gives ½CV²) or V = Q/C (gives ½Q²/C).
- Energy ∝ V²: **double the voltage, four times the energy**.

Why it is ½QV and not QV

The supply does work **QV** pushing charge Q through p.d. V. Only **half** ends up stored in the capacitor; the other half is dissipated as heat in the resistance of the circuit while it charges.

WORKED EXAMPLE

energy stored and released

A 2200 μF capacitor is charged to 12 V.

W = ½CV² = 0.5 × 2200 × 10^{-6} × 12^2 = **0.16 J**

It discharges to 6.0 V. Energy released = ½C($V_1^2 - V_2^2$) = 0.5 × 2.2 × 10^{-3} × (144 − 36) = **0.12 J**

The p.d. halved but **three-quarters** of the energy was released.

With a +20% tolerance on C, the maximum energy at 12 V is 1.2 × 0.158 = 0.19 J.

WORKED EXAMPLE

A 220 μF capacitor is charged to a p.d. of 9.0 V. Calculate (a) the charge stored and (b) the energy stored. (c) The p.d. is increased to 18 V. How much extra energy is stored?

Show solution

(a) $Q = CV = 220 \times 10^{-6} \times 9.0 = 1.98 \times 10^{-3}\ \text{C} \approx 2.0\ \text{mC}$

(b) $W = \frac{1}{2}CV^2 = \frac{1}{2} \times 220 \times 10^{-6} \times 9.0^2 = 8.9 \times 10^{-3}\ \text{J}$

(c) Energy goes as V^2, so doubling V gives four times the energy: $4 \times 8.91 \times 10^{-3} = 3.56 \times 10^{-2}\ \text{J}$. The extra energy is $3.56 \times 10^{-2} - 0.89 \times 10^{-2} = 2.7 \times 10^{-2}\ \text{J}$ (three times the original, not one).

WORKED EXAMPLE

A camera flash uses a 150 μF capacitor charged to 300 V. When the flash fires, 90% of the stored energy is released in 1.0 ms. Calculate the mean power delivered to the flash tube, and explain why the camera cannot simply use its 3 V battery directly.

Show solution

Energy stored: $W = \frac{1}{2}CV^2 = \frac{1}{2} \times 150 \times 10^{-6} \times 300^2 = 6.75\ \text{J}$

Energy released: $0.90 \times 6.75 = 6.08\ \text{J}$, so mean power $= 6.08 / 1.0 \times 10^{-3} = 6.1 \times 10^3\ \text{W}$

The battery has enough energy, but its internal resistance limits the current it can deliver, so it can only supply energy slowly (a few watts). The capacitor is charged slowly from the battery over a few seconds, then delivers the energy very quickly through the low resistance of the flash tube (small RC).

MARK-SCHEME WORDING

- "Use of W = ½CV²" (or ½QV or ½Q²/C).
- "Use of tolerance as +20%" is a separate mark: take the extreme value in the direction the question needs.
- Comparisons need a conclusion: "184 J > 150 J, so enough energy".

Often asked as: energy stored; energy released between two voltages; tolerance questions; comparing with another energy store (battery, spring); MCQ on how W changes when V changes.

COMMON MISTAKE

Forgetting to square V.

Leaving V in mV or C in μF.

Taking energy released as $\frac{1}{2}C(V_1 - V_2)^2$. It is $\frac{1}{2}C(V_1^2 - V_2^2)$: the **difference of two energies**.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

The battery does work QV while the capacitor stores only ½QV (WPH14 Oct 2021 MCQ).

SPEC 102 Charging and discharging curves; the time constant RC

WHAT THIS SPEC POINT ASKS

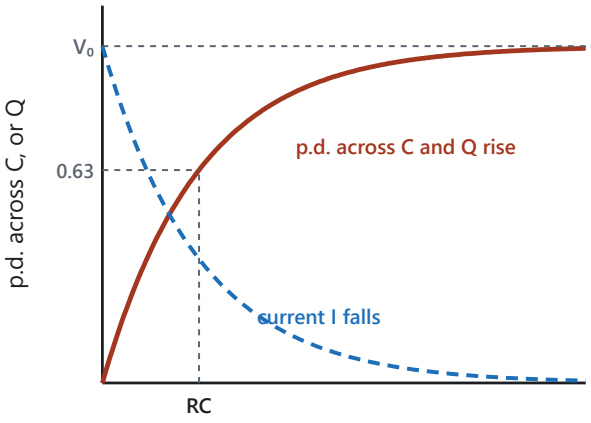
Draw and interpret charge and discharge curves for RC circuits, and understand the significance of the time constant RC.

Charging through a resistor (from a supply V_0)

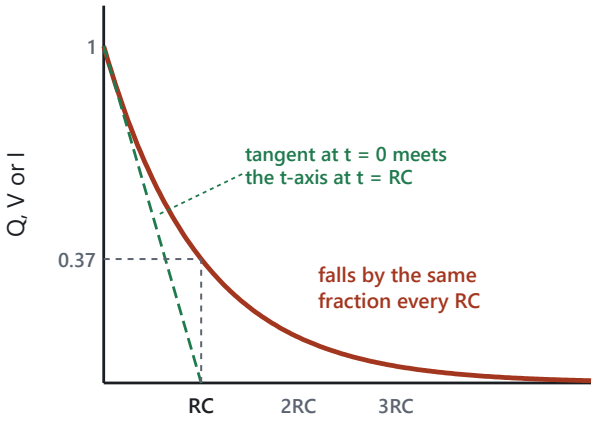
1. At the start the capacitor has no charge, so $V_C = 0$, all of V_0 is across R, and the current is **maximum**: $I_0 = V_0/R$.
2. Charge builds up on the plates, so the **p.d. across C rises**.
3. Since $V_R + V_C = V_0$, the **p.d. across R falls**, so the **current falls** ($I = V_R/R$).
4. Charge therefore arrives more slowly: Q and V_C rise more and more slowly.
5. Finally $V_C = V_0$ and **I = 0**.

Discharging through a resistor

- At the start $V = V_0$ and $I_0 = V_0/R$. As charge flows off, V falls, so I falls, so the charge flows off more slowly.
- Q, V and I **all** decay **exponentially**: each falls by the same fraction in each equal time interval.



Charging through R from a supply V_0



Discharging through R (fraction of start value)

Left: charging. The p.d. across C (and Q) rise to V_0, reaching 63% after one time constant; the current (dashed, on its own scale) falls from I_0. Right: discharging. Q, V and I all fall to 37% of their start value after RC; the tangent at t = 0 cuts the time axis at t = RC.

The time constant τ = RC

Time constant

$$\tau = RC$$

τ: time constant / s R: resistance / Ω C: capacitance / F

Half-life of a discharging capacitor

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 After one half-life, the charge is half its starting value:

$$\tfrac{1}{2}Q_0 = Q_0\, e^{-t_{1/2}/RC} \quad \Rightarrow \quad \tfrac{1}{2} = e^{-t_{1/2}/RC}$$

2 Take natural logs of both sides. $\ln \tfrac{1}{2} = -\ln 2$:

$$-\ln 2 = -\frac{t_{1/2}}{RC} \quad \Rightarrow \quad \boxed{t_{1/2} = RC \ln 2 \approx 0.69\, RC}$$

The starting charge cancelled out, which is why the half-life is the same however full the capacitor is.

- τ = time constant (s), R = resistance (Ω), C = capacitance (F). $\Omega \times F = (V\,A^{-1})(C\,V^{-1}) = C\,A^{-1} = s$.
- **Discharging:** after one time constant Q, V and I have fallen to **1/e ≈ 37%** of their initial values.
- **Charging:** after one time constant Q and V_C have risen to **63%** of their final values.
- After about **5RC** the capacitor is effectively fully charged or discharged (less than 1% left).
- Time to halve: $t_{½}$ = RC ln 2 ≈ **0.69RC**.
- Bigger R or bigger C → bigger τ → **slower** charging and discharging.

WORKED EXAMPLE

time constant

R = 47 kΩ, C = 220 μF. $\tau = RC = 47 \times 10^{3} \times 220 \times 10^{-6}$ = **10 s**.

Time to fall to half: $t_{½}$ = 0.69 × 10.3 = **7.2 s**.

MARK-SCHEME WORDING

Six-mark charging explanation (indicative content):

- "There is a **current** in the circuit; the **charge on the capacitor increases**."
- "The **p.d. across the capacitor increases**, so the **p.d. across the resistor decreases**."
- "So the **current decreases** (I = V/R) and the rate of charging falls."
- "Finally the p.d. across C equals the supply p.d. and the current is zero."

Sketches: "exponential curve from the origin levelling at V_0", "passes about 2/3 of the maximum at t = RC", "levels off after 4–5 time constants"; discharge "starts on the y-axis" and "does not cross or crash into the t-axis". Label the axes with quantities and units.

Often asked as: sketch a charging or discharging graph of V or I; explain the shape; effect of a changed resistance on the time constant; where $V_R = V_C$ (each half the supply).

COMMON MISTAKE

Explaining with "charges moving" but never mentioning **current** and **p.d.**: no marks.

Using 37% on a **charging** curve (it is 63% for charging).

Saying the effect of a change without going through the time constant or current first.

Curves that hit the time axis.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Components whose resistance changes with p.d. (an LED or diode in the discharge path): as V falls, R rises, so the time constant **increases** (WPH14 Jan 2020).
- A capacitor as a **backup supply**: its p.d. "remains high enough to operate the device for a short time" (WPH14 Oct 2021).

SPEC 103 Core Practical 11: capacitor charge and discharge

WHAT THIS SPEC POINT ASKS

Use an oscilloscope or data logger to display and analyse the p.d. across a capacitor as it charges and discharges through a resistor.

CORE PRACTICAL

Core practical 11: method

Circuit: d.c. supply (e.g. 6.0 V), two-way switch, resistor R (e.g. 100 kΩ) and capacitor C (e.g. 220–470 μF) in series. A **voltmeter, data logger or oscilloscope** goes **in parallel with the capacitor**. In one switch position the supply charges C through R; in the other, C discharges through R alone.

1. Set the data logger to record V against time at a suitable rate (several readings per second for τ of tens of seconds).
2. Switch to charge. Record V until it stops changing (about 5RC).
3. Switch to discharge and record V as it falls.
4. Repeat for other values of R (or C).

Oscilloscope version: replace the supply and switch with a **square-wave signal generator**, so the capacitor charges and discharges repeatedly. Adjust the time-base so one charge–discharge cycle fills the screen, then read V and t from the grid. The period of the square wave should be much longer than RC (about 10RC).

Safety: connect an **electrolytic** capacitor the right way round (+ to +) and do not exceed its working voltage, or it can overheat and burst.

Analysis

- From $V = V_0 e^{-t/RC}$, taking natural logs: **$\ln V = \ln V_0 - t/RC$**.
- Plot **ln V** (y) against **t** (x): a straight line with **gradient −1/RC** and intercept $\ln V_0$.
- So RC = −1/gradient, and C = −1/(R × gradient).
- Quick checks: the time to fall to $0.37V_0$ (discharge) or rise to $0.63V_0$ (charge) is RC.

WORKED EXAMPLE

capacitance from a ln V graph

R = 100 kΩ. The best-fit line on a graph of ln V (V in volts) against t (in s) passes through (0, 1.79) and (40, 0.46).

Gradient = (0.46 − 1.79) ÷ 40 = −0.0333 s^{-1}

RC = 1 ÷ 0.0333 = 30.1 s, so C = 30.1 ÷ 1.00 × 10^5 = **3.0 × 10^{-4} F** (300 μF)

Errors and improvements

- Capacitors have large **tolerances** (often ±10% or ±20%), so compare your C with the nominal value using the tolerance.
- A voltmeter with too low a resistance gives a second discharge path and makes the discharge look faster. A data logger or oscilloscope has a very high resistance.
- Reading by eye with a stopwatch is limited by reaction time: choose RC of at least tens of seconds, or use a data logger.
- Use the best-fit line over many points rather than one pair of readings.

WORKED EXAMPLE

In Core Practical 11 a student discharges a capacitor through a 47 kΩ resistor and plots $\ln(V/\mathrm{V})$ against t. The line passes through (0 s, 1.79) and (60 s, 0.29). Determine the capacitance, and the p.d. the capacitor was charged to.

Show solution

Gradient $= \dfrac{0.29 - 1.79}{60 - 0} = -0.0250\ \mathrm{s}^{-1}$

Gradient $= -1/RC$, so $RC = 1/0.0250 = 40\ \mathrm{s}$, and $C = \dfrac{40}{47 \times 10^3} = 8.5 \times 10^{-4}\ \mathrm{F} = 850\ \mu\mathrm{F}$

Intercept $= \ln V_0 = 1.79$, so $V_0 = e^{1.79} = 6.0\ \mathrm{V}$

MARK-SCHEME WORDING

- Circuit: "battery, resistor and capacitor in **series**", "voltmeter / data logger **in parallel with the capacitor**", "switch arranged to give a **discharge path** through the resistor".
- Why a given circuit is wrong: "when charging the voltmeter is not across C" or "when discharging the resistor is not in the circuit".
- Tolerance: "6640 Ω is within 5% of 6800 Ω" with the calculation shown.

Often asked as: draw the circuit; read V and t off a data-logger or oscilloscope trace; find R or C from it; check a component is within tolerance.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

A repeating charge–discharge trace: energy released per discharge = ½C($V_1^2 - V_2^2$) (WPH14 Jun 2025).

SPEC 104 Exponential discharge equations

WHAT THIS SPEC POINT ASKS

Use Q = $Q_0e^{(-t/RC)}$, derive and use I = $I_0e^{(-t/RC)}$ and V = $V_0e^{(-t/RC)}$, and the matching log equations.

Capacitor discharge

FORMULA SHEET

$$Q = Q_0\, e^{-t/RC}$$
$$I = I_0\, e^{-t/RC}$$
$$V = V_0\, e^{-t/RC}$$

Q_0, I_0, V_0: values at $t = 0$ t: time since the discharge began / s RC: time constant / s

Deriving V = $V_0e^{(-t/RC)}$ and I = $I_0e^{(-t/RC)}$

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Start from the discharge equation and divide both sides by C. Since $V = Q/C$ at every instant:

$$\frac{Q}{C} = \frac{Q_0}{C}\, e^{-t/RC} \quad \Rightarrow \quad \boxed{V = V_0\, e^{-t/RC}}$$

$V_0 = Q_0/C$ is the p.d. at the start. C is a constant, so dividing by it doesn't change the shape.

2 During discharge the capacitor is connected directly across the resistor, so the p.d. across the resistor equals the p.d. across the capacitor. Ohm's law gives the current:

$$I = \frac{V}{R} = \frac{V_0}{R}\, e^{-t/RC}$$

3 V_0/R is the current at the start, I_0, so:

$$\boxed{I = I_0\, e^{-t/RC}}$$

- Q_0, I_0, V_0 = values at t = 0; t = time since discharge began (s); RC = time constant (s).
- **Deriving the V form:** V = Q/C, and C is constant, so $V = Q_0e^{-t/RC}/C = V_0e^{-t/RC}$.
- **Deriving the I form:** during discharge the p.d. across R equals V, so $I = V/R = (V_0/R)e^{-t/RC} = I_0e^{-t/RC}$.

Log forms

Log forms

FORMULA SHEET

$$\ln Q = \ln Q_0 - \frac{t}{RC}$$
$$\ln I = \ln I_0 - \frac{t}{RC}$$
$$\ln V = \ln V_0 - \frac{t}{RC}$$

plot $\ln V$ (or $\ln Q$, $\ln I$) against t: gradient $= -1/RC$

Deriving the log equations

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1 Take natural logs of both sides of $Q = Q_0e^{-t/RC}$:

$$\ln Q = \ln\left(Q_0\, e^{-t/RC}\right)$$

2 The log of a product is the sum of the logs, $\ln(ab) = \ln a + \ln b$:

$$\ln Q = \ln Q_0 + \ln\left(e^{-t/RC}\right)$$

3 $\ln$ undoes e, so $\ln(e^x) = x$:

$$\boxed{\ln Q = \ln Q_0 - \frac{t}{RC}}$$

4 Exactly the same steps on the p.d. and current equations give:

$$\boxed{\ln V = \ln V_0 - \frac{t}{RC}} \qquad \boxed{\ln I = \ln I_0 - \frac{t}{RC}}$$

Compare with $y = c + mx$: plotting $\ln V$ against t gives a straight line with gradient $-1/RC$ and intercept $\ln V_0$.

- Each has the form y = c + mx: plot ln Q (or ln I, ln V) against t to get a straight line of **gradient −1/RC**.
- Rearranged to find a time: **t = RC ln(V₀/V)**.

What fraction is left?

Time	1RC	2RC	3RC	4RC	5RC
Fraction left (discharge), e^{-n}	0.37	0.14	0.050	0.018	0.0067
Fraction reached (charging), $1 - e^{-n}$	0.63	0.86	0.95	0.98	0.99

WORKED EXAMPLE

current and time during discharge

C = 100 μF charged to 12 V discharges through R = 2.0 kΩ. RC = 2.0 × 10³ × 100 × 10⁻⁶ = 0.20 s; I₀ = 12 ÷ 2000 = 6.0 mA.

Current after 0.30 s: I = 6.0 × $e^{-0.30/0.20}$ = 6.0 × 0.223 = **1.3 mA**

Time for V to fall to 3.0 V: t = RC ln(V₀/V) = 0.20 × ln(12/3.0) = 0.20 × 1.386 = **0.28 s**

Charging equations

- During charging the p.d. across the **resistor** decays: $V_R = V_0 e^{-t/RC}$, and so does the current.
- Because $V_C = V_0 - V_R$:

Capacitor charging

$$V_C = V_0\left(1 - e^{-t/RC}\right)$$
$$Q = Q_0\left(1 - e^{-t/RC}\right)$$

V_0: supply p.d. / V Q_0: final charge, CV_0 / C t: time since charging began / s

- Example: the same circuit charged from 12 V reaches 9.0 V when $e^{-t/0.20} = 0.25$, so t = 0.20 × ln 4 = 0.28 s.

Ways to find RC from a graph (all credited)

1. **Time-constant route:** read the time when V = 0.37V_0 (discharge) or 0.63V_0 (charge).
2. **Tangent route:** draw the tangent at t = 0; it meets the t-axis (or the final-value line) at t = RC.
3. **Substitution route:** take a pair of (t, V) values and solve $V = V_0e^{-t/RC}$ using logs.
4. **Half-life route:** read $t_{½}$, then RC = $t_{½}$/ln 2.
5. **Log-graph route:** gradient of ln V against t = −1/RC.

WORKED EXAMPLE

A 470 μF capacitor is charged to 12.0 V and then discharged through a 2.2 kΩ resistor. Calculate (a) the time constant, (b) the initial current, (c) the p.d. after 2.0 s, (d) the time for the p.d. to fall to 3.0 V and (e) the percentage of the original energy still stored at that moment.

Show solution

(a) $RC = 2200 \times 470 \times 10^{-6} = 1.034\ \text{s} \approx 1.03\ \text{s}$

(b) $I_0 = V_0/R = 12.0/2200 = 5.45 \times 10^{-3}\ \text{A} = 5.45\ \text{mA}$

(c) $t/RC = 2.0/1.034 = 1.934$, so $V = 12.0\,e^{-1.934} = 12.0 \times 0.1445 = 1.73\ \text{V}$

(d) $t = RC\ln(V_0/V) = 1.034 \times \ln(12.0/3.0) = 1.034 \times 1.386 = 1.43\ \text{s}$

(e) $W \propto V^2$, so the fraction left is $(3.0/12.0)^2 = 1/16 = 6.25\%$. Three quarters of the p.d. has gone, but nearly 94% of the energy has gone.

MARK-SCHEME WORDING

- "Use of $V = V_0e^{-t/RC}$ **Or** use of ln V = ln V_0 − t/RC", "use of time constant = RC".
- "Takes corresponding pairs of values of V and t from the graph."

- Choosing a standard capacitor: pick the value that keeps the time **safe** (e.g. calculated 0.56 F → choose 0.58 F so the time is not too short).
- Answers read from graphs are marked within a range.

Often asked as: C or R from a graph; time to reach a given p.d. or current; derive the charging equation; percentage left after n time constants; tolerance checks.

COMMON MISTAKE

Putting values from a **charging** graph straight into the **decay** equation. Use $V_0 - V_C$ (the resistor's p.d.) or the $(1 - e^{-t/RC})$ form.

Taking V_0 as the highest reading (5.8 V) instead of the supply p.d. (6.0 V).

Finding "the time for V to fall by a factor of e" when RC itself was asked for, and vice versa: read the question.

Using "fully charged after 5RC" as a calculation method.

No unit on the final answer.

Quick checks

QUICK CHECK

A charged capacitor discharges through a resistor. What fraction of its original charge is left after a time 2RC?

- 0.25
- 0.37
- 0.14
- 0.63

$Q/Q_0 = e^{-2} = 0.135$. Each time constant multiplies the charge by 0.37, so two of them give $0.37^2 \approx 0.14$. 0.25 would be two **half-lives**, and a half-life is shorter than RC.

QUICK CHECK

The p.d. across a capacitor is doubled. What happens to the energy stored?

- It stays the same
- It doubles
- It increases by a factor of 4
- It halves

$W = \frac{1}{2}CV^2$ and C is fixed, so $W \propto V^2$. Doubling V also doubles Q, and $W = \frac{1}{2}QV$ gives the same factor of 4.

QUICK CHECK

Which of these is equivalent to the unit of the time constant, Ω × F?

- A s^{-1}
- s
- C V
- J s^{-1}

$\Omega = \mathrm{V\ A^{-1}}$ and $\mathrm{F = C\ V^{-1}}$, so $\Omega\,\mathrm{F = C\ A^{-1} = (A\ s)\ A^{-1} = s}$.

QUICK CHECK

An uncharged capacitor is connected in series with a resistor and a battery. At the instant the switch is closed, which statement is correct?

- The current is zero because the capacitor has no charge.
- The current is at its maximum, V_0/R. ✓
- The p.d. across the capacitor is V_0.
- The p.d. across the resistor is zero.

With no charge on it, the capacitor has no p.d. across it, so the whole e.m.f. is across the resistor. The current is V_0/R, its largest value. It falls as V_C builds up.

QUICK CHECK

A student plots ln I against t for a capacitor discharging through a resistor. What is the gradient of the line?

- −RC
- ln I_0
- −1/RC ✓
- 1/RC

$\ln I = \ln I_0 - t/RC$ has the form $y = c + mx$ with $m = -1/RC$. The intercept is $\ln I_0$.

QUICK CHECK

A capacitor is charged to a fixed p.d. and discharged through a resistor. The resistor is replaced with one of half the resistance and the experiment is repeated. Which statement is correct?

- The initial current halves and the total charge that flows doubles.
- The time constant doubles.
- **The initial current doubles and the total charge that flows is unchanged.**
- The energy dissipated in the resistor halves.

$I_0 = V_0/R$ doubles and RC halves, so the discharge is twice as fast. But the capacitor started with the same $Q_0 = CV_0$ and energy $\frac{1}{2}CV_0^2$, and all of it still flows through the resistor (the area under the I–t graph is unchanged).